

Trading Strategy Research on BTC and ETH

How we moved from statistics to strategies, with honest results (4-hour bars, 2021–2025)

1. Setup

- **Data:** Binance BTC/USDT and ETH/USDT, 4-hour candles, Jan 2021 to Dec 2025. Checks found no zero-volume bars and no invalid OHLC bars.
- **Why 4-hour bars:** daily bars give too few data points. 1-hour bars give too many bars, more noise and more trading cost. 4-hour bars (6 per day) are a good middle choice for testing a strategy. We did not run a separate timeframe comparison; this was a design choice.
- **Backtest rules:** start capital 100,000; cost 0.15% on every trade; a signal is acted on in the *next* bar (no look-ahead). Sharpe is annualised with 6×365 bars per year.
- **Benchmarks:** buy & hold, and random signals. Random signals lose everything (about –100%), which shows that costs are a real hurdle.
- **Which data was used for what:** all research and tuning used **2021–2023**. GARCH parameters were fitted on data before 2023 and tested from 2023. The pairs strategy was tested from 2022. The years **2024–2025** were used only at the end, with all parameters frozen, so they are the only truly out-of-sample years for the trend strategies. Note that 2023 is *not* out-of-sample for them, because their parameters were tuned using data up to 2023.

2. First idea: mean reversion

We first checked the autocorrelation of 4-hour returns. At lag 1 it is negative: **–0.034 for BTC and –0.038 for ETH**. The 95% band is about ± 0.025 , so the value is outside the band, but it is very small. A negative value means a big move tends to be followed by a small move the other way, so we tried a mean reversion strategy.

Strategy: take the z-score of the latest return against the last 14 bars. Go against the move when $|z| > 2$, exit when z returns to 0, and stop out at $|z| = 4$.

2021–2023	Return	Sharpe	Max drawdown	Trades	Win rate
BTC mean reversion	–83.9%	–2.15	–88.6%	558	43.7%
ETH mean reversion	–80.1%	–1.39	–86.0%	559	46.0%
BTC buy & hold	+44.0%	0.51	–77.0%	1	–
ETH buy & hold	+205.4%	0.86	–81.1%	1	–

Result: bad. The strategy trades very often (average hold about 0.27 days). For BTC, gross profit was 174k and gross loss 258k. The small edge in the autocorrelation was not big enough to pay the 0.15% cost on each trade. **Lesson:** a statistically visible pattern is not always a tradeable one.

3. Variance ratio: a hint towards trend

The variance ratio (VR) compares the variance of multi-bar returns with the variance of single-bar returns. VR above 1 suggests trending; below 1 suggests mean reversion.

- **BTC:** VR is above 1 only for lags of about 190 to 570 bars (roughly 30 to 95 days), with a peak near 1.08. At short lags it sits around 0.95–0.98, and it falls to about 0.75 at lag 1000.
- **ETH:** VR never goes above 1 (peak about 0.99, then it falls to about 0.65).

So BTC shows a mild trend at medium horizons, and this pointed us towards trend-following strategies. This is only a hint: we did not run a significance test on the VR.

4. ADX: when is the market trending?

ADX measures how strong a trend is (0 to 100), not its direction. A common rule is that ADX above 25 means a real trend. The ADX plot shows that the market spends most of its time below 25, with bursts above 25 (often 30–55). So a trend strategy should trade only in those bursts. We used **ADX > 25** as a filter in all trend strategies. (The notebook plot uses a 30-bar ADX to look smoother; the strategies use the standard 14. The spike to 100 at the start of the plot is only a warm-up effect.)

5. Trend strategies

MA crossover, version 1 (simple)

Go long when the 30-bar moving average is above the 90-bar one, go short when it is below, but only when ADX > 25. There is no stop loss. This was the first, basic version.

MA crossover, version 2 (with stop loss)

Version 2 enters on a fresh crossover (with ADX > 25), long or short, and adds a **dynamic ATR stop loss** (explained below). Parameters were found by repeated trial and error:

- BTC: fast 50, slow 200, ATR period 25, $k = 5.5$
- ETH: fast 60, slow 230, ATR period 14, $k = 6$

Dynamic stop loss

The stop is a **trailing ATR stop**. For a long trade: stop = highest high since entry – $k \times \text{ATR}$. It only moves up, never down. ATR (average true range) measures recent volatility, so the stop is wider when the market is wild and tighter when it is calm. With k between 5.5 and 7 the stop is wide and gives the trend room to breathe.

2021–2023	BTC return	BTC Sharpe	ETH return	ETH Sharpe
Buy & hold	+44.0%	0.51	+205.4%	0.86
Random	–100%	–5.80	–100%	–3.93
MA crossover v1	–47.5%	–0.20	+10.7%	0.37
MA crossover v2	–16.6%	0.20	+17.9%	0.46

Result: v1 loses money on BTC. v2 is better, but still clearly behind buy & hold. We also noticed in trial runs that the plain MA crossover (v1) is **not robust**: a small change in the parameters caused big changes in the result (this was an observation from trial runs; the notebook has no grid for v1). The results were still not good.

MA cross long (final trend strategy)

We then **removed the short signals** and **added a cooldown** after every exit. The rules:

- Enter long when fast MA > slow MA and ADX > 25, and the cooldown is over.
- Exit when the ATR trailing stop is hit *or* the trend flips (fast MA < slow MA). Then wait 6–7 bars (about one day) before trading again.
- BTC: fast 40, slow 200, ATR period 20, $k = 7$, cooldown 7. ETH: fast 60, slow 230, ATR period 14, $k = 6$, cooldown 6. These were again found by trial and error.

2021–2023	Return	Sharpe	Max drawdown	Trades	Win rate
BTC ma_cross_long	+93.9%	0.77	–53.8%	34	35.3%
BTC buy & hold	+44.0%	0.51	–77.0%	1	–
ETH ma_cross_long	+151.6%	0.85	–57.8%	38	44.7%
ETH buy & hold	+205.4%	0.86	–81.1%	1	–

The win rate is low (about 35% for BTC), but the average win (about 19k) is three times the average loss (about 6k). This is typical for trend following: a few big winners pay for many small losers.

Robustness check

We ran each strategy on a grid of 25 combinations (fast MA 30–70, slow MA 150–260, other parameters fixed) over 2021–2023 and looked at the Sharpe ratio:

Strategy / asset	Lowest Sharpe	Median	Highest	Share of grid with Sharpe > 0
MA crossover v2, BTC	–0.32	0.08	0.49	68%
MA crossover v2, ETH	–0.05	0.37	0.65	96%
MA cross long, BTC	0.15	0.71	1.09	100%
MA cross long, ETH	0.37	0.62	0.90	100%

- **BTC:** v2 is sensitive (a small parameter change can flip the result). **ma_cross_long** is robust and was the best BTC strategy in the 2021–2023 period.
- **ETH:** both strategies are almost robust, but neither gave a clearly better Sharpe or profit than buy & hold in 2021–2023.

6. Final results with frozen parameters (2021–2025)

We ran all strategies on the full history with the same parameters. Only 2024–2025 is new data.

2021–2025	BTC return	BTC Sharpe	BTC max DD	ETH return	ETH Sharpe	ETH max DD
Buy & hold	+198.5%	0.67	–77.0%	+297.8%	0.75	–81.1%
Mean reversion	–94.4%	–2.23	–96.2%	–95.9%	–1.72	–97.4%
MA crossover v1	–47.1%	–0.06	–65.2%	+40.8%	0.41	–69.0%
MA crossover v2	–45.1%	0.06	–70.7%	+169.7%	0.64	–78.9%
MA cross long	+83.2%	0.52	–53.8%	+361.2%	0.88	–57.8%

Honest reading:

- **BTC:** buy & hold beats ma_cross_long on return and Sharpe over 2021–2025. The strategy's real benefit is **protection in falling markets** (max drawdown –54% vs –77%). Its lead over buy & hold in 2021–2023 (section 5) came from the tuning period only.
- **ETH:** over 2021–2025 ma_cross_long beats buy & hold on return (+361% vs +298%), Sharpe (0.88 vs 0.75) and drawdown (–58% vs –81%). In 2021–2023 alone it did not. We report both.

Share of quarters beating buy & hold

The organisers rank strategies by how many quarters they beat the benchmark, so we computed it (quarters beaten / total quarters):

Strategy	BTC 2021–23	BTC 2021–25	BTC 2024–25	ETH 2021–23	ETH 2021–25	ETH 2024–25
Mean reversion	2/12	3/20	1/8	4/12	6/20	2/8
MA crossover v1	3/12	4/20	1/8	4/12	8/20	4/8
MA crossover v2	3/12	5/20	2/8	4/12	7/20	3/8
MA cross long	6/12	7/20	1/8	5/12	9/20	4/8

In the out-of-sample years 2024–2025, `ma_cross_long` beat BTC in only 1 of 8 quarters (BTC was in a strong bull run), and 4 of 8 for ETH (average excess return about –1% per quarter). No strategy reliably beats buy & hold quarter by quarter.

7. Statistical models: GARCH volatility

Idea: we cannot predict the *direction* of returns, but maybe we can predict how *big* the moves will be.

- Raw returns show almost no autocorrelation (values about ± 0.05 at most).
- **Squared returns** do: autocorrelation is 0.06–0.20 for BTC and 0.13–0.23 for ETH over 15 lags, far above the ± 0.025 band. This is called **volatility clustering** (also called an ARCH effect): calm periods follow calm periods, and wild periods follow wild periods. Absolute returns are used later, in the forecast check.
- This led us to **GARCH(1,1)** with Student-t errors. Results: BTC $\alpha = 0.039$, $\beta = 0.961$; ETH $\alpha = 0.046$, $\beta = 0.954$. Since $\alpha + \beta$ is about 1, volatility shocks are very persistent. The t-distribution parameter v is 3.1 (BTC) and 3.5 (ETH), which means fat tails.
- **Limit of the model:** the Ljung-Box test on squared standardized residuals still rejects (BTC $p = 0.0006$, ETH $p = 0.03$ at lag 10). So GARCH(1,1) does not capture all of the pattern.
- **Forecast check:** forecast volatility has a correlation of about 0.20 with the realised absolute return. Realised absolute return rises steadily across the five forecast-volatility groups (BTC 0.39% to 0.78%; ETH 0.37% to 0.82%). So the forecast ranks risk correctly.

How we used it

We scaled the position of `ma_cross_long`: size = target volatility / forecast volatility, capped at 1 and rounded to steps of 0.25. The target is the median volatility of the training period. GARCH parameters were fitted on data before 2023 and re-fitted every 90 days using only past data, so all GARCH tests start in 2023. **`garch_v2`** uses a rolling 2-year window and adds a kill switch (go flat when forecast volatility is in the top 10% of the last 90 days).

2023–2025	BTC return	BTC Sharpe	BTC max DD	ETH return	ETH Sharpe	ETH max DD
Buy & hold	+428.6%	1.43	–34.4%	+148.1%	0.80	–65.1%
Plain <code>ma_cross_long</code>	+74.7%	0.75	–43.7%	+133.2%	0.92	–40.4%
With GARCH sizing	+70.5%	0.73	–44.7%	+115.7%	0.87	–40.6%
<code>garch_v2</code>	+66.9%	0.76	–34.0%	+56.1%	0.60	–38.9%

Result: GARCH sizing did not help. The position size was 1.0 about 96% of the time for BTC and 99% for ETH, so the GARCH version is almost the same as the plain one (BTC 2023 Sharpe 2.07 vs 2.10). A likely reason is that the target volatility, taken from the high-volatility 2021–2022 period, is rarely exceeded; we did not test this. `garch_v2` cut the BTC drawdown (–34% vs –44%) with about the same Sharpe, but it hurt ETH (Sharpe 0.60 vs 0.92). GARCH forecasts volatility well, but this did not turn into a better trading result here.

8. Cointegration: ETH/BTC pairs

Idea: ETH and BTC move together (return correlation **0.85**). If their prices are cointegrated, the spread between them should return to its mean, and we could trade that. But high correlation does not prove cointegration, so we ran the Engle-Granger test (ETH on BTC, daily log prices):

Sample	Days	Hedge ratio	EG statistic	5% critical	p-value	Half-life (days)
Full 2021–23	1095	0.81	−3.31	−3.34	0.054	42.9
2021-06 onwards	944	1.00	−2.45	−3.34	0.302	48.4
2022	365	1.03	−2.15	−3.35	0.450	28.6
2023	365	0.56	−2.88	−3.35	0.141	12.7

What the metrics say: the test does **not** support cointegration. The full sample is close to the 5% line but still fails ($p = 0.054$), and every shorter sample fails clearly. The hedge ratio is unstable (0.56 to 1.03). Half-lives of 13–48 days are slow for a 4-hour strategy. Because the evidence was weak, we expected this strategy to be weak too.

We still tested it, to see what happens. The strategy trades the z-score of the spread (entry at 2, exit at 0, stop at 4). The hedge ratio is re-fitted every 90 days on the past year.

What is the “gate”?

The gate is a rolling cointegration filter. At every re-fit, we run the Engle-Granger test on the past year of data. If $p < 0.10$, the gate is **on** and the strategy may trade. If not, it stays flat. **No gate** means the filter is switched off, so the strategy always trades. The gate was on only about 25% of the time in 2022–2023 and 12% over 2022–2025.

ETH/BTC spread	Return	Sharpe	Max drawdown	Trades
Gate, 2022–23	+3.2%	0.30	−5.9%	1
No gate, 2022–23	−5.6%	−0.29	−16.7%	5
Gate, 2022–25	+3.2%	0.22	−5.9%	1
No gate, 2022–25	−15.7%	−0.13	−37.5%	12

Result: the gate version looks better only because it almost never trades. One winning trade is not evidence of an edge. Without the gate the strategy loses money. So **pairs trading is not supported by our tests**, and we do not use it. (One small inconsistency: the Engle-Granger table uses daily prices, while the gate inside the strategy uses 4-hour prices.)

9. Summary and limits

- Mean reversion looked reasonable in the statistics but lost to trading costs.
- Trend following worked better. The best strategy was **ma_cross_long** (long only, ADX filter, ATR trailing stop, cooldown). It is robust to small parameter changes, and its main benefit is smaller drawdowns.
- On BTC it beat buy & hold only in the tuning period; on the full period and in 2024–2025 buy & hold was better. On ETH it was better over 2021–2025.
- GARCH measures volatility well but did not improve the strategy. Cointegration was not supported, so pairs trading was not pursued.

Limits to keep in mind:

- Parameters were tuned by trial and error on 2021–2023, so those results are in-sample.

- Only two assets and about five years of data. Trend strategies make few trades (34–68 for `ma_cross_long`), so results can change a lot with luck.
- No significance tests on the variance ratio or on the strategy results. Costs are a flat 0.15%; real slippage was not modelled.
- **Other metrics used:** Sharpe, Sortino, max drawdown, win rate, average win/loss, holding time, and share of quarters beating the benchmark.